

# *AML & ALL Classification via analysis of Gene Expression*

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## 1. Background

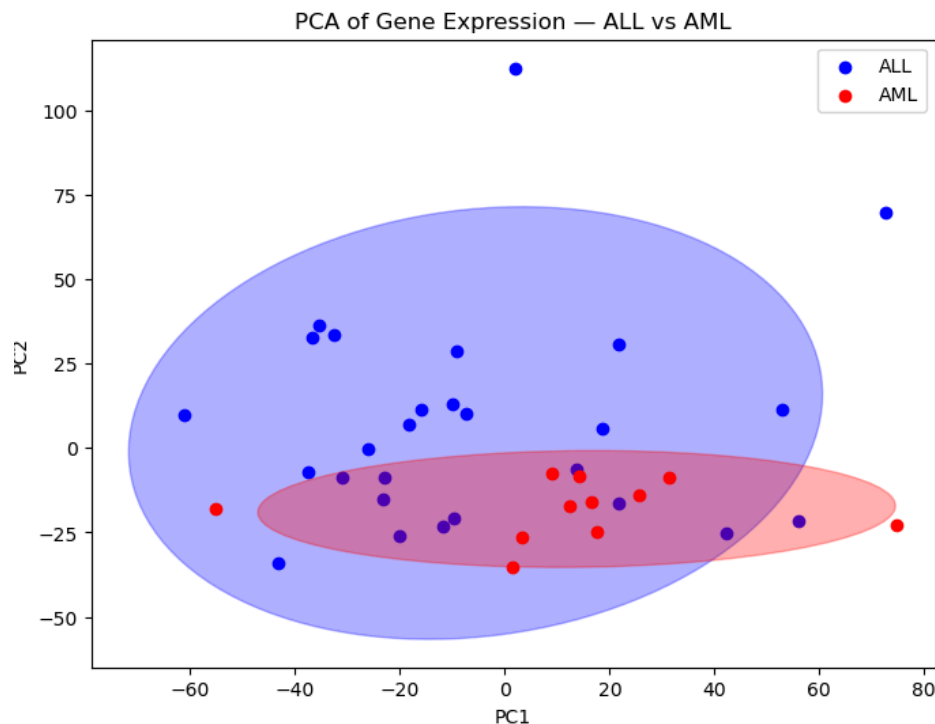
The task set out for this project was to replicate a study done in the late 90s that used gene expression analysis to determine the subset of leukaemia in a dataset of patients, at the time this was quite revolutionary for the field of computational biology. The paper recorded an achieved accuracy of 85% however in my project I only achieved 80%.

**ALL (Acute Lymphoblastic Leukaemia)** - *a cancer of the lymphoid cells, most commonly affecting children.*

**AML (Acute Myeloid Leukaemia)** - *a cancer of the myeloid cells, more commonly affecting adults.*

Prior to 1999, leukaemia subtype classification relied on manual microscopic examination of blood cells; this would obviously be fallible due to human error, it was time-consuming and was occasionally inconclusive (doctors disagreeing with each other's assessments). [Golub et al. \(1999\)](#) paper addressed this by demonstrating that gene expression profiling could objectively classify cancer subtypes computationally without prior biological knowledge of the patient. Using 72 leukaemia patients, they showed that *Acute Lymphoblastic Leukaemia (ALL)* and *Acute Myeloid Leukaemia (AML)* exhibit distinctly different gene expression profiles, and that a machine learning classifier could accurately distinguish between them based solely on gene activity levels. Published in Science with over 8,000 citations, it remains one of the most influential papers in computational biology and helped establish the foundation for modern precision medicine.

This project replicates that work using modern Python tools and accessible libraries, using the same dataset generated during the study, provided from [kaggle.com](https://www.kaggle.com).

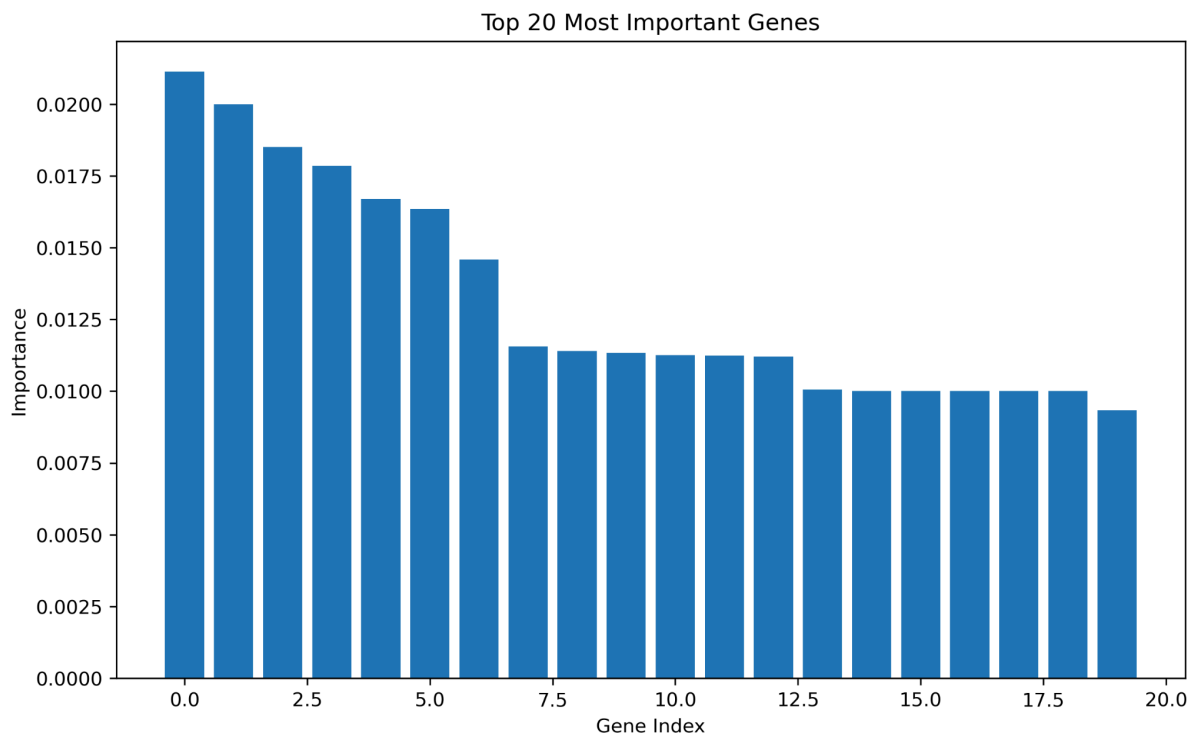


## 2. Results

- 2.1 Data Overview & Preprocessing
  - *The dataset consists of gene expression measurements from 72 leukaemia patients across 7,129 genes.*
  - *Preprocessing involved :*
    - *removing non-informative call columns,*
    - *transposing the matrix so that rows represent patients and columns represent genes, and normalising the data using StandardScaler to ensure all gene expression values were on a comparable scale.*
    - *Removed AFFX genes from the dataset - as these were synthetic sequences and not real - existed solely for control.*
    - *We split dataset 80% for training and 20% for testing - this was because I was getting an accuracy of circa 64% likely due to how little AML training data was in it with my original split (11 total records) - this new split gave more training data. Now 57 patients records are for training with 15 for testing.*
- 2.2 Unsupervised Analysis — PCA
  - Principal Component Analysis (PCA) was applied to reduce the 7,129 gene dimensions to two principal components for visualisation. The resulting scatter plot shows a degree of separation between ALL and AML patients along PC1, suggesting that gene expression patterns alone carry enough signal to distinguish the two subtypes without using class labels.
- 2.3 Classification — Random Forest
  - A Random Forest classifier was trained on the gene expression data to predict leukaemia subtype. The model initially achieved an accuracy of 64.7% on the test set with my original split - later I re-examined the dataset and instead split the data by

80% for training and 20% for testing, after training following we achieved an accuracy of 80%. The confusion matrix revealed the model performed well on ALL but struggled with AML, likely due to class imbalance in the training data — 27 ALL patients versus only 11 AML patients. I returned later to properly split the training and test data to attempt to achieve a higher accuracy.

- 2.4 Classification — SVM
  - A Support Vector Machine with a linear kernel was also trained as a comparison. It achieved a similar accuracy of 80% but showed a more balanced distribution of errors across both classes, correctly identifying more AML patients at the cost of some ALL misclassifications.
- 2.5 Feature Importance — Most Informative Genes
  - Random Forest feature importance scores were used to identify the genes most informative for distinguishing ALL from AML. The top genes show a clear drop-off in importance, with a small number of genes accounting for a disproportionate share of the classification signal.
  - Most common Genes are **S81439\_at**, **X91220\_at**, **M33680\_at**.

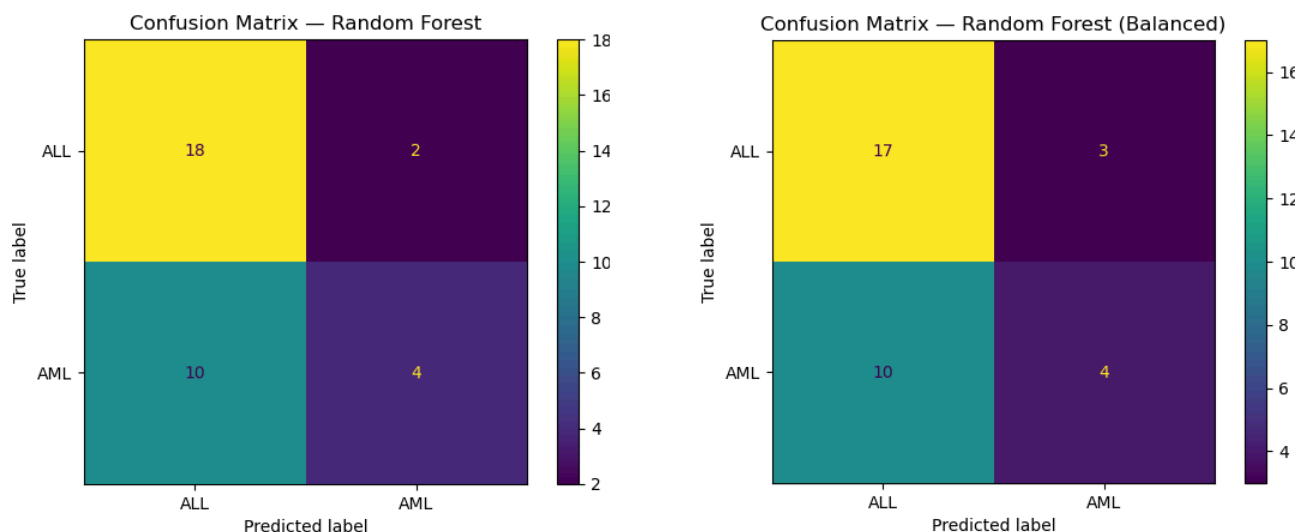


### 3. Conclusion

- 3.1 Answering the Research Question
  - Replicating this work proved that classification between these subsets of leukaemia by gene expression is possible, as documented by this scientific paper.
  - The first time I attempted to develop this script in Python, I was experiencing particularly low accuracy, circa 60%, which obviously is unacceptable in a binary classification task. Reviewing the code and properly splitting the training and test of the dataset at the start of the script, this helped improve accuracy; it was just an imbalance of AML data.

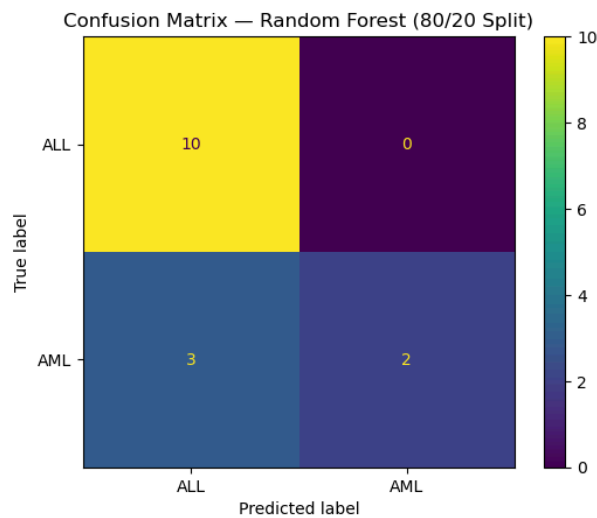
- 3.2 Limitations
  - Several limitations should be noted.
    - First, the dataset is small — only 72 patients — which restricts the model's ability to generalise, particularly for the minority AML class, which had only 11 training examples. This class imbalance likely explains the poor AML classification performance observed in both models.
    - Second, the pre-defined train/test split of 38/34 is suboptimal; a standard 80/20 stratified split or k-fold cross-validation would likely yield more robust results. Third, the feature importance analysis initially identified AFFX control probes — synthetic microarray calibration sequences — as the most informative features, highlighting the importance of careful data cleaning in omics analysis.
    - Finally, while gene expression alone shows promising discriminatory power as demonstrated by the PCA separation, a more comprehensive clinical model would integrate additional data types such as cytogenetics, patient age, and white blood cell counts.
- 3.3 Future Work
  - Address **class imbalance** using oversampling techniques such as SMOTE (Synthetic Minority Oversampling Technique) to generate synthetic AML training examples
  - **Hyperparameter tuning** via grid search to optimise Random Forest and SVM performance
  - Integrate additional **clinical data types** such as cytogenetics, patient age, and white blood cell counts for a more comprehensive multi-modal model
  - Extend the analysis to **multi-class classification** beyond ALL vs AML to include leukaemia subtypes such as T-cell ALL vs B-cell ALL
  - My project achieved an accuracy of 80% with the Golub et al.'s original achieving 85%.

*Model's performance before making improvements - Before changes accuracy on models was 60%.*

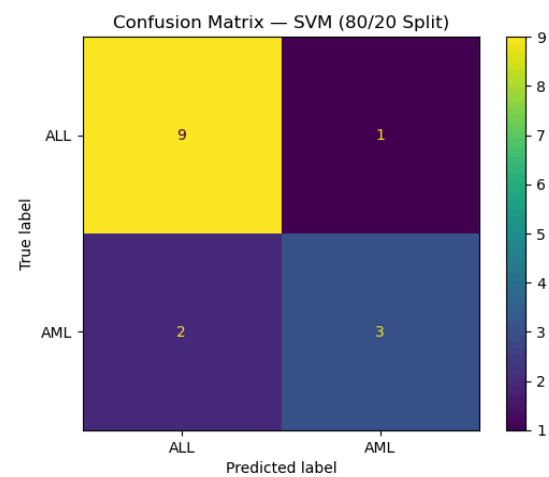


*Accuracy improved from 64.7% to 80% following implementation of a stratified 80/20 train/test split, increasing AML training examples from 11 to 20 and addressing the class imbalance issue identified in the initial model.*

Accuracy: 0.8



Accuracy: 0.8



## 4. References

- [Golub, T.R., Slonim, D.K., Tamayo, P., Huard, C., Gaasenbeek, M., Mesirov, J.P., Coller, H., Loh, M.L., Downing, J.R. and Lander, E.S. \(1999\) 'Molecular classification of cancer: Class discovery and class prediction by gene expression monitoring', \*Science\*, 286\(5439\), pp. 531–537. doi: \[10.1126/science.286.5439.531\]\(https://doi.org/10.1126/science.286.5439.531\)](#)

Crawford, J. (2017) *Gene expression dataset (Golub et al.)* [Dataset]. Kaggle. Available at: <https://www.kaggle.com/datasets/crawford/gene-expression> (Accessed: June 2026)